

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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Problem Statement: #55

BOOK CLUB READING CHALLENGE & DISCUSSION PORTAL

Functional requirements

FR-001 — Track Reading Goals

The system shall allow a club member to set and track personal reading page goals.

FR-002 — Manage Chapter Discussions

The system shall allow club members to participate in chapter-based discussion threads.

FR-003 — Protect Spoiler Content

The system shall hide spoiler-tagged discussion content until the member explicitly reveals it or has marked the relevant chapter as read.

FR-004 — Conduct Monthly Polls

The system shall allow members to vote in monthly polls for selecting the next book.

FR-005 — Manage Discussion Content

The system shall allow the Discussion Lead to manage chapter discussion threads and monthly book-selection polls.

Non-Functional Requirements

NFR-001 — Security & Performance

The system shall prevent duplicate votes using user session verification and update monthly poll tallies in real time.

NFR-002 — Performance

The system shall display reading-goal and discussion information within an acceptable response time during normal and peak usage.

Main Challenges

1. Security

Spoiler content must not accidentally be exposed before the member is permitted to view it.

2. Performance

Monthly poll results need to update in real time, and discussion pages should load efficiently.

3. Usability

Members should easily:

- Track reading progress
- Access chapter discussions
- Reveal permitted spoilers
- Vote in monthly polls

4. Data consistency

Voting must prevent duplicate votes and maintain accurate poll results.

Architectural Style Analysis

A. Layered Architecture

Structure

The system is divided into layers such as:

- Presentation Layer
- Business Logic Layer
- Data Layer

Advantages

- Clear separation of concerns
- Easy to understand
- Easier maintenance
- Suitable for traditional web applications

Disadvantages

- Some performance overhead
- Tight coupling between layers
- Difficult to scale individual components

B. Microservices Architecture

Structure

- The system consists of small independently deployable services.

Advantages

- Independent scaling
- Fault isolation
- Independent deployment
- Technology diversity

Disadvantages

- Operational complexity
- Network latency
- Data consistency challenges.

C. Client – Server Architecture

Structure

- Clients communicate with a centralized server.

Advantages

- Centralized control
- Simple deployment
- Easy data consistency

Disadvantages

- Single point of failure
- Scalability bottleneck
- Limited fault tolerance.

ARCHITECTURE SELECTION AND JUSTIFICATION

We chose Layered Architecture for the Book Club Reading Challenge & Discussion Portal.

Architectural Choice:

Layered Architecture divides the system into Presentation, Business Logic and Data layers. This provides a clear separation between user interaction, application functionality and data management.

Reason 1:

The portal contains distinct functionalities such as reading goal tracking, chapter discussions, spoiler handling and monthly voting. Separating these responsibilities into business components makes the system easier to understand, maintain and modify.

Reason 2:

A centralized Data Layer provides consistent storage and retrieval of member information, reading goals, discussion content and voting information. This is particularly useful for maintaining accurate poll results and reading progress.

Security Advantage:

The User & Session Manager can perform session verification and access control. This helps prevent duplicate votes and protects spoiler-tagged discussion content from being revealed before the member is permitted to view it.

Performance Benefit:

Layered Architecture separates reading goals, discussions and voting into distinct business components, allowing each function to be processed

efficiently. The Voting & Poll Manager can support real-time poll tally updates, while the separation of responsibilities helps maintain responsive access to reading-goal and discussion information during normal and peak usage.

COMPONENTS

Component 1 — User Interface

Handles:

- Reading dashboard
- Reading goals
- Discussion threads
- Polls
- User interactions

Component 2 — User & Session Manager

Handles:

- User sessions
- Member authentication
- Session verification
- Access control

Component 3 — Reading Goal Manager

Handles:

- Personal reading goals
- Pages read
- Reading progress
- Goal tracking

Component 4 — Discussion Manager

Handles:

- Chapter discussion threads
- Discussion comments
- Spoiler-tagged content
- Spoiler visibility

Component 5 — Voting & Poll Manager

Handles:

- Monthly polls
- Book-selection voting
- Duplicate-vote prevention
- Real-time vote tally updates

Component 6 — Book Club Database

Stores:

- Member information
- Reading goals
- Reading progress
- Discussion threads
- Comments
- Spoiler information
- Polls
- Votes
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INTERFACES

No.	Components	Interface
1	User Interface ↔ User & Session Manager	Authentication & Session API
2	User Interface ↔ Reading Goal Manager	Reading Goal API
3	User Interface ↔ Discussion Manager	Discussion API
4	User Interface ↔ Voting & Poll Manager	Voting API
5	Business Components ↔ Database	Data Access Interface

COMPONENT DIAGRAM

